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Kwon et al.

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(54) **WASHING MACHINE**

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Feb. 9, 2021 (KR) 10-2021-0018038

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D06F 39/08 (2006.01)

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CPC ***D06F 34/04*** (2020.02); ***D06F 39/085*** (2013.01)

(58) **Field of Classification Search**
CPC D06F 34/04; D06F 39/085; D06F 34/28; D06F 39/08; D06F 39/12; D06F 39/081
USPC 68/208
See application file for complete search history.

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(57) **ABSTRACT**

A washing machine including a cabinet, a tub disposed inside the cabinet, a drain pump configured to discharge water from the tub, a service port to be connectable to a service device and configured to receive or transmit error information of the washing machine therethrough when the service device is connected to the service port, and a pump housing mounted on a part of the cabinet in which the drain pump is located, wherein the service port is disposed inside the pump housing.

13 Claims, 14 Drawing Sheets

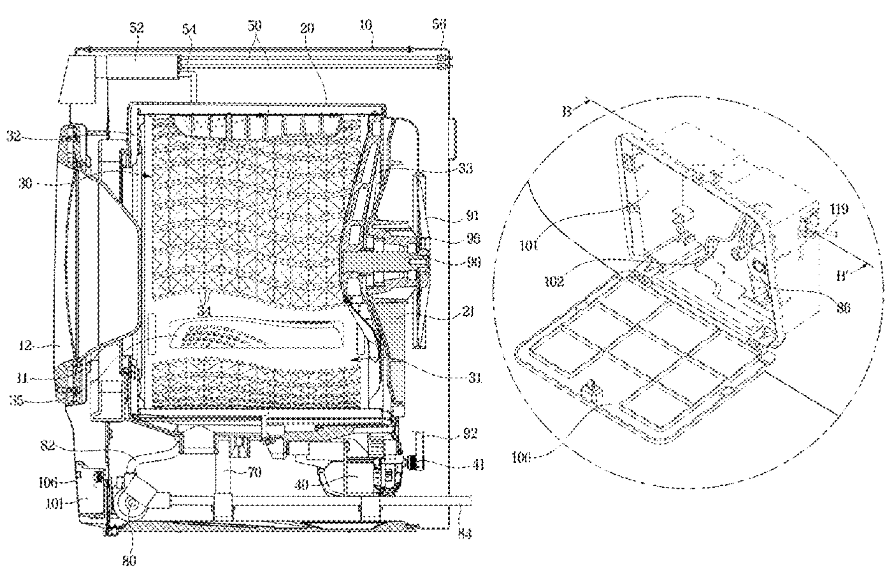

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FIG. 1

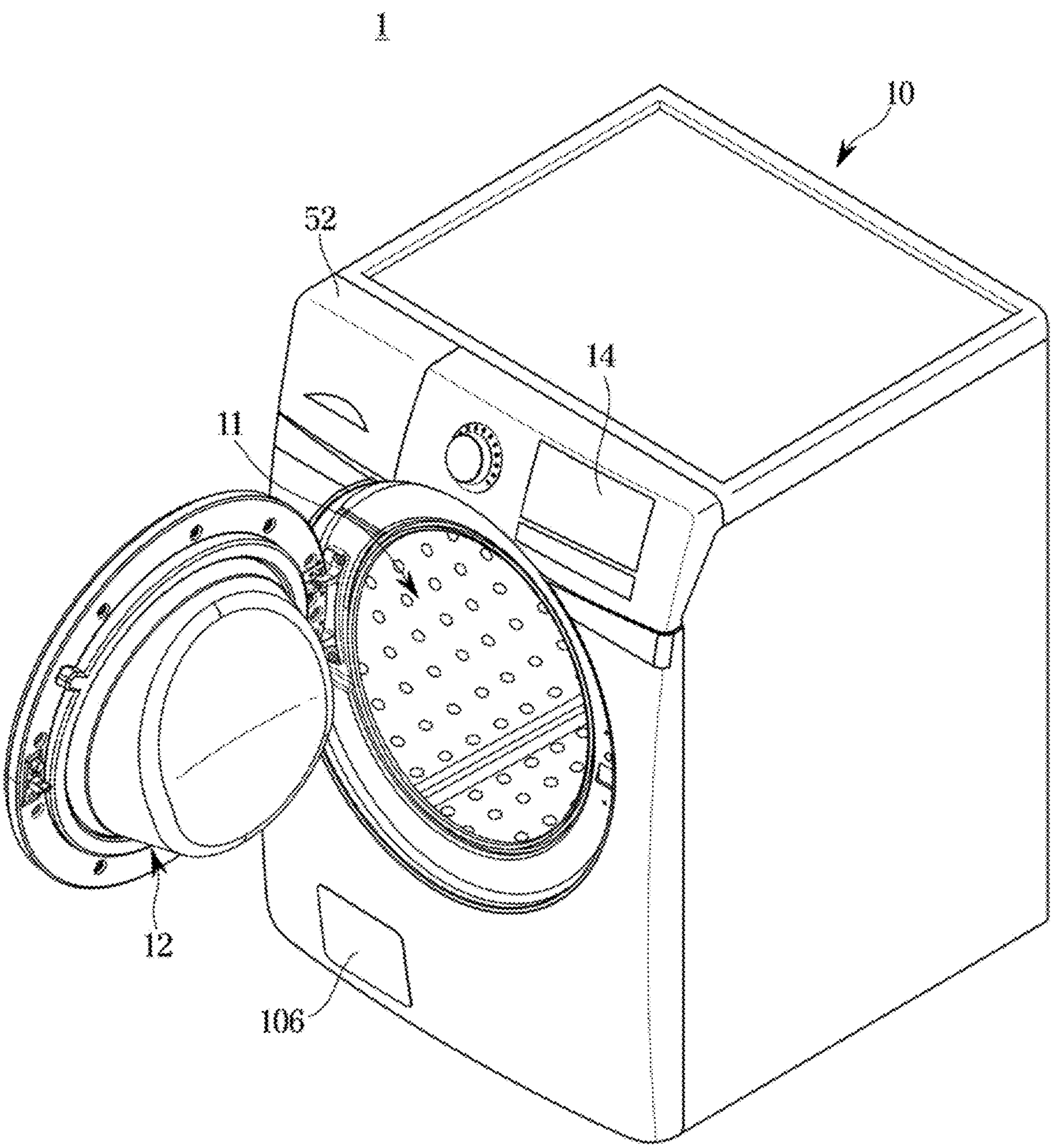

FIG. 2

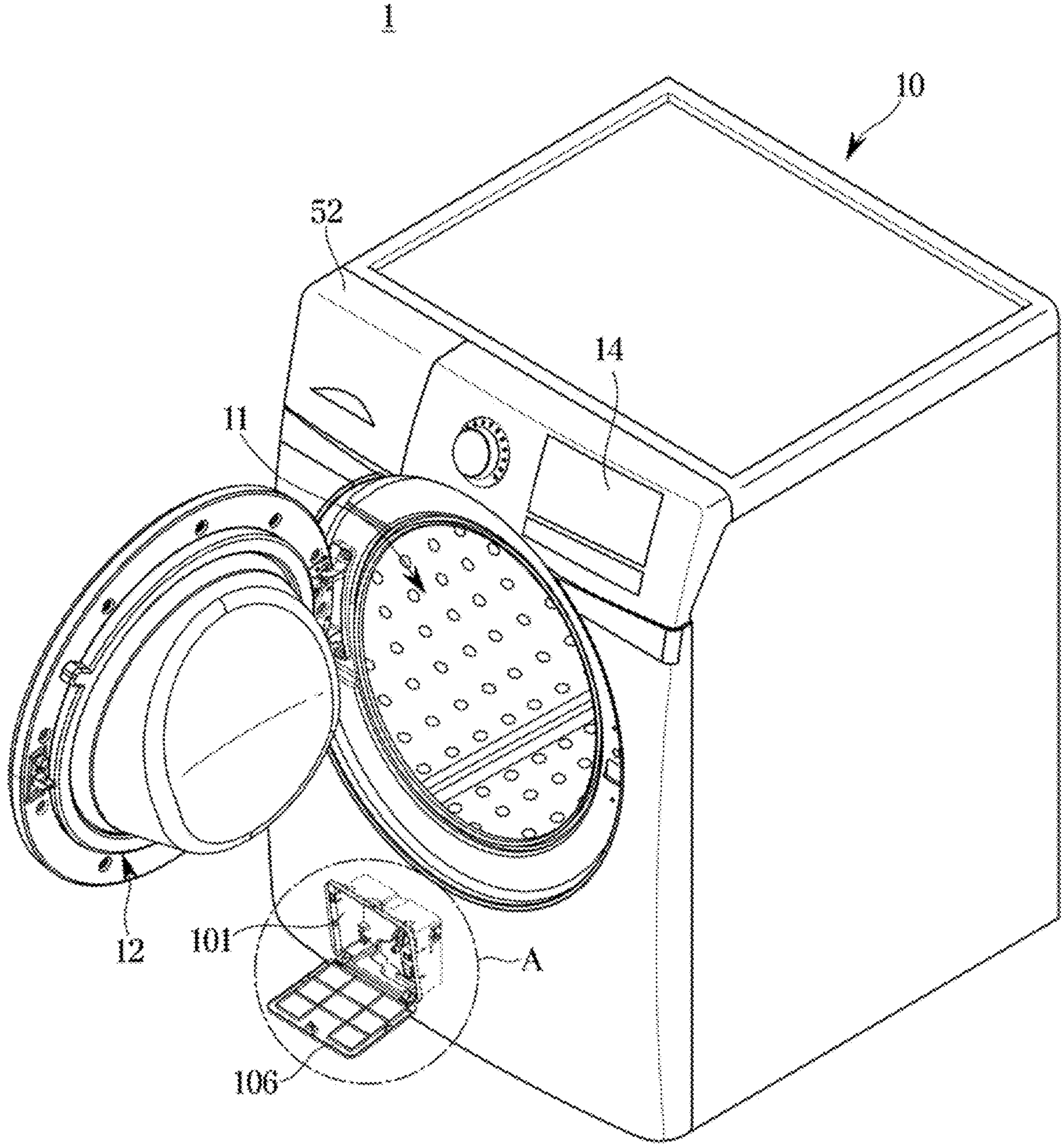

FIG. 3

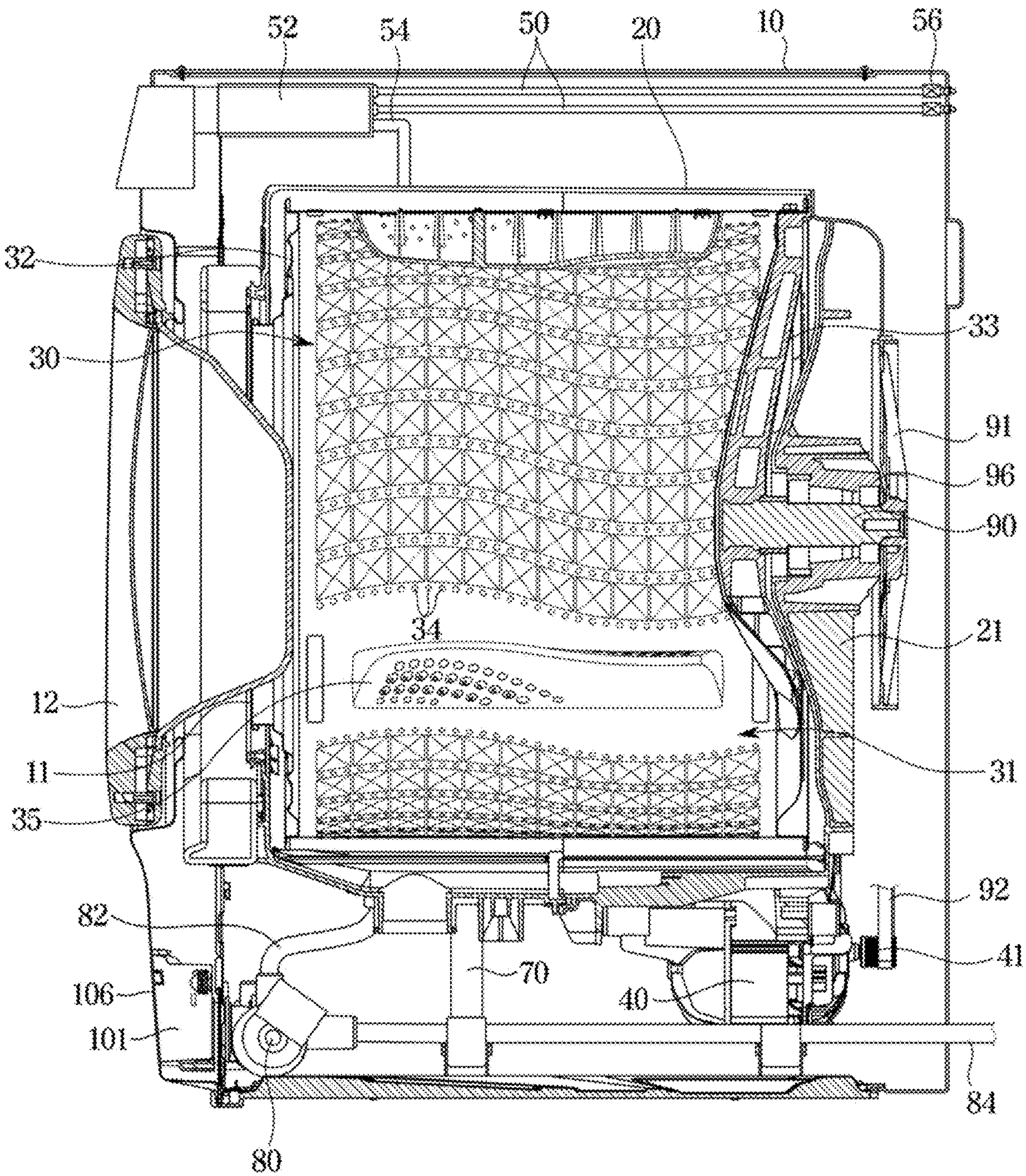
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FIG. 4

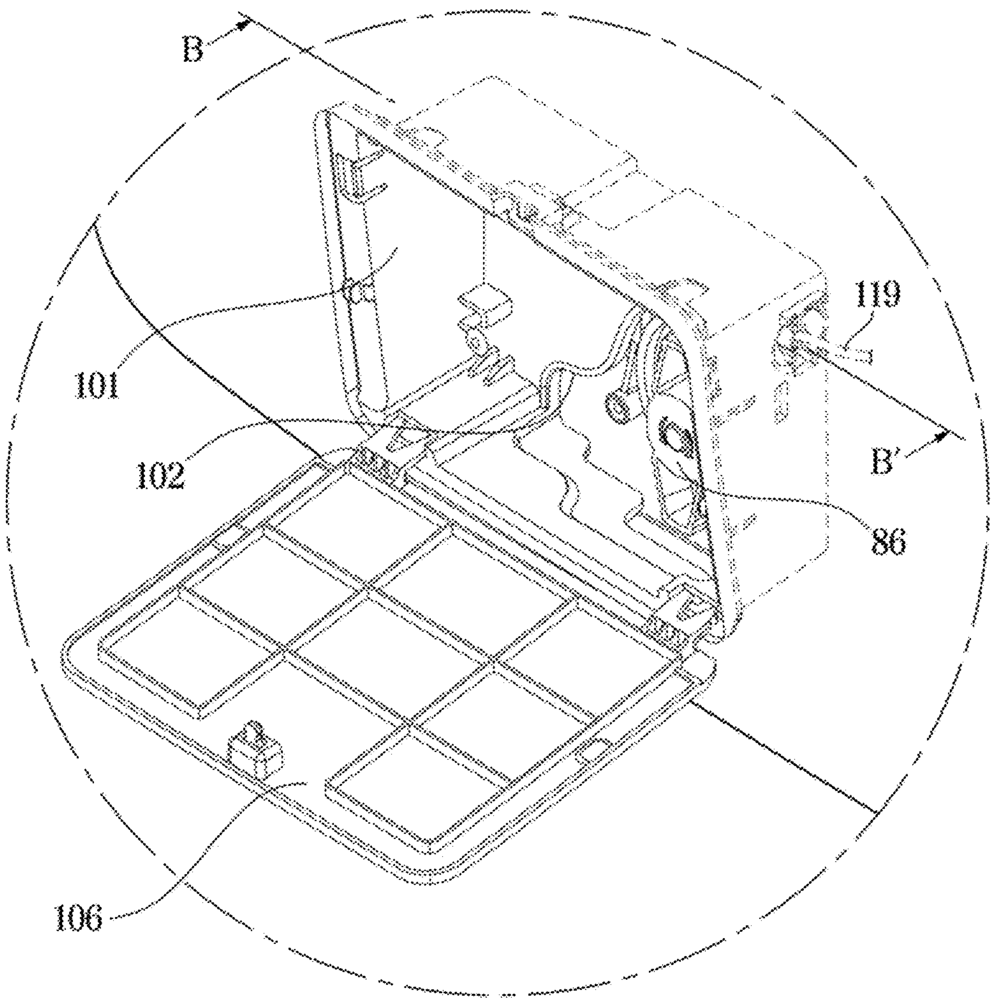


FIG. 5

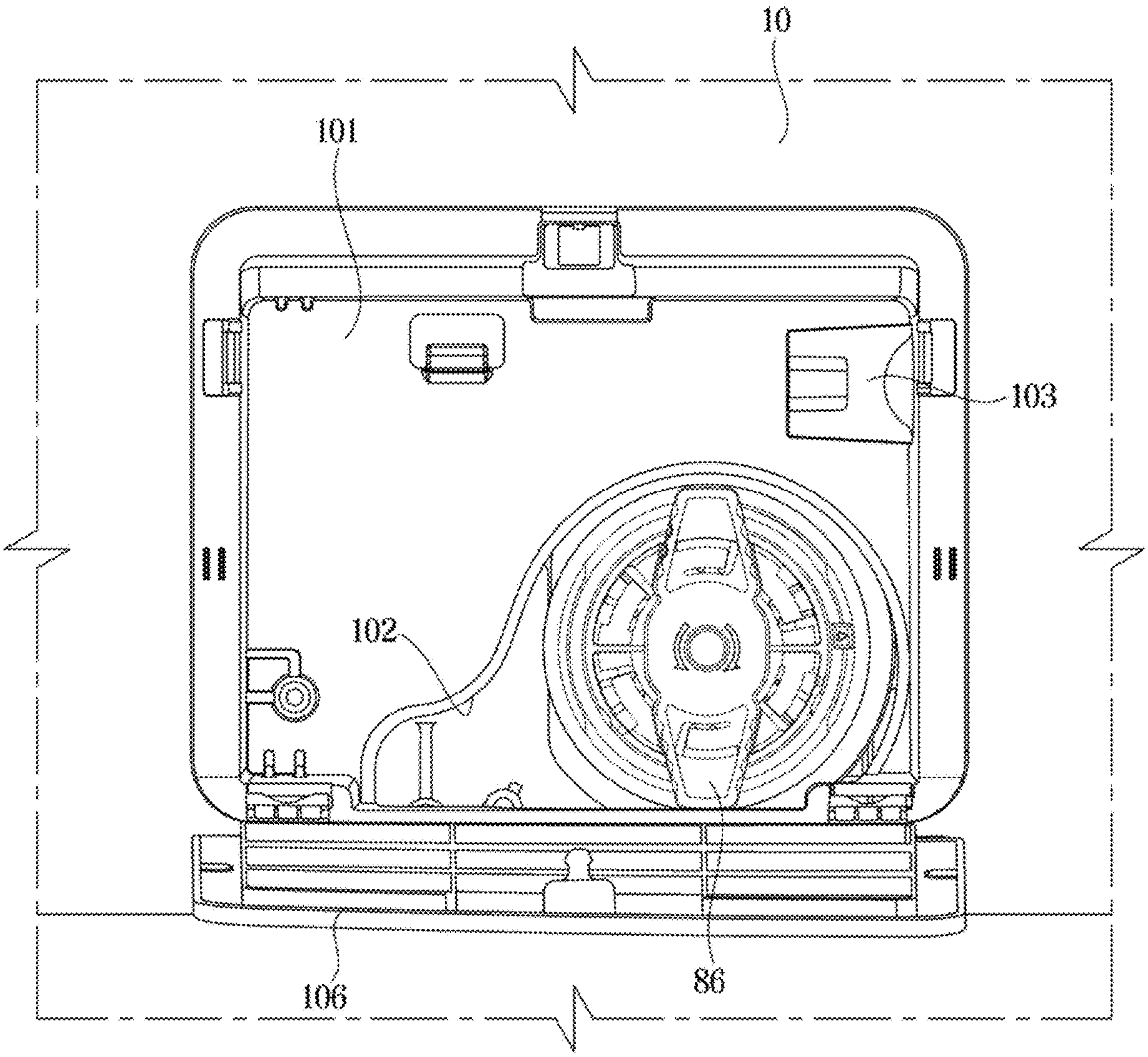


FIG. 6

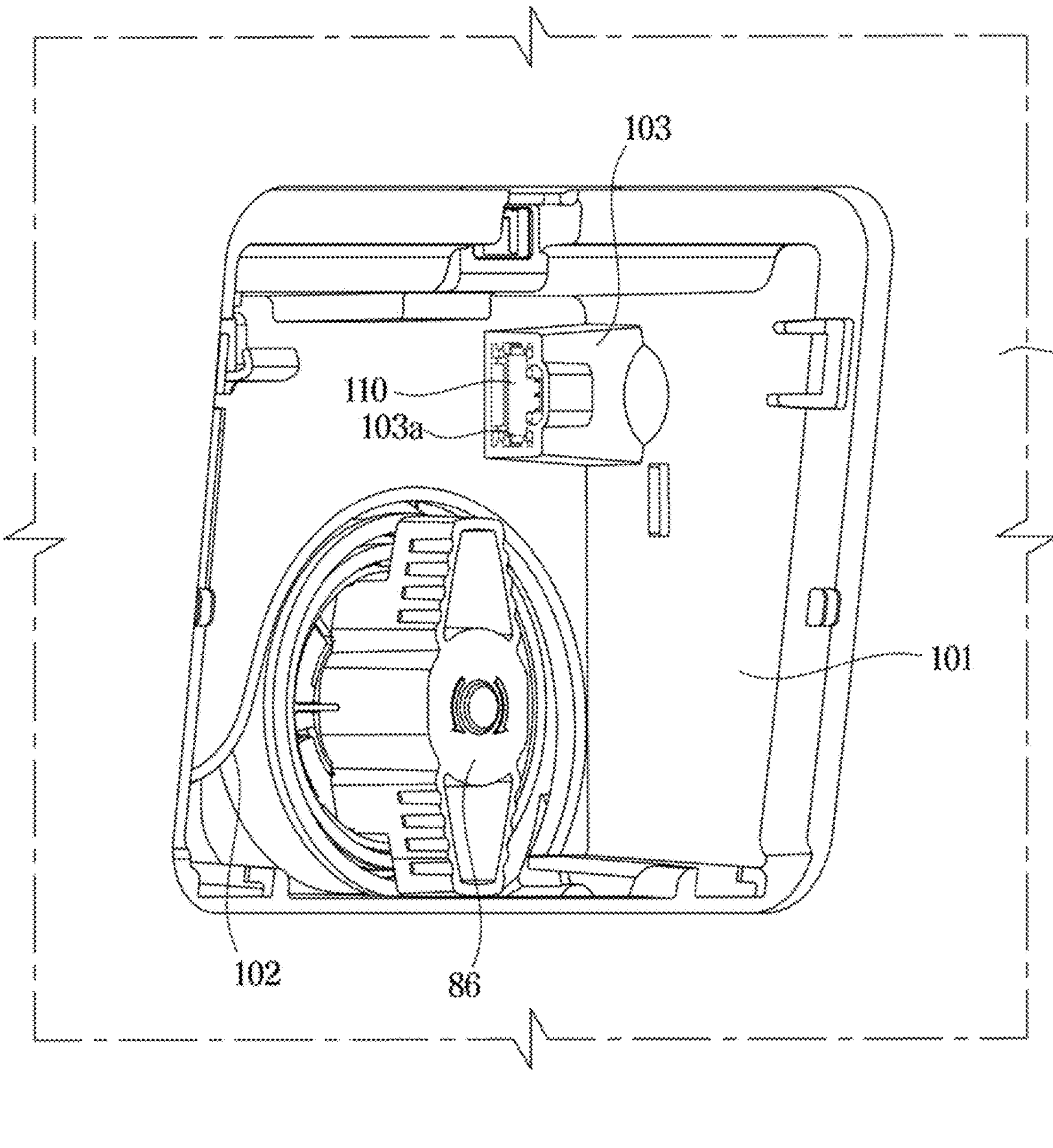


FIG. 7

FIG. 8

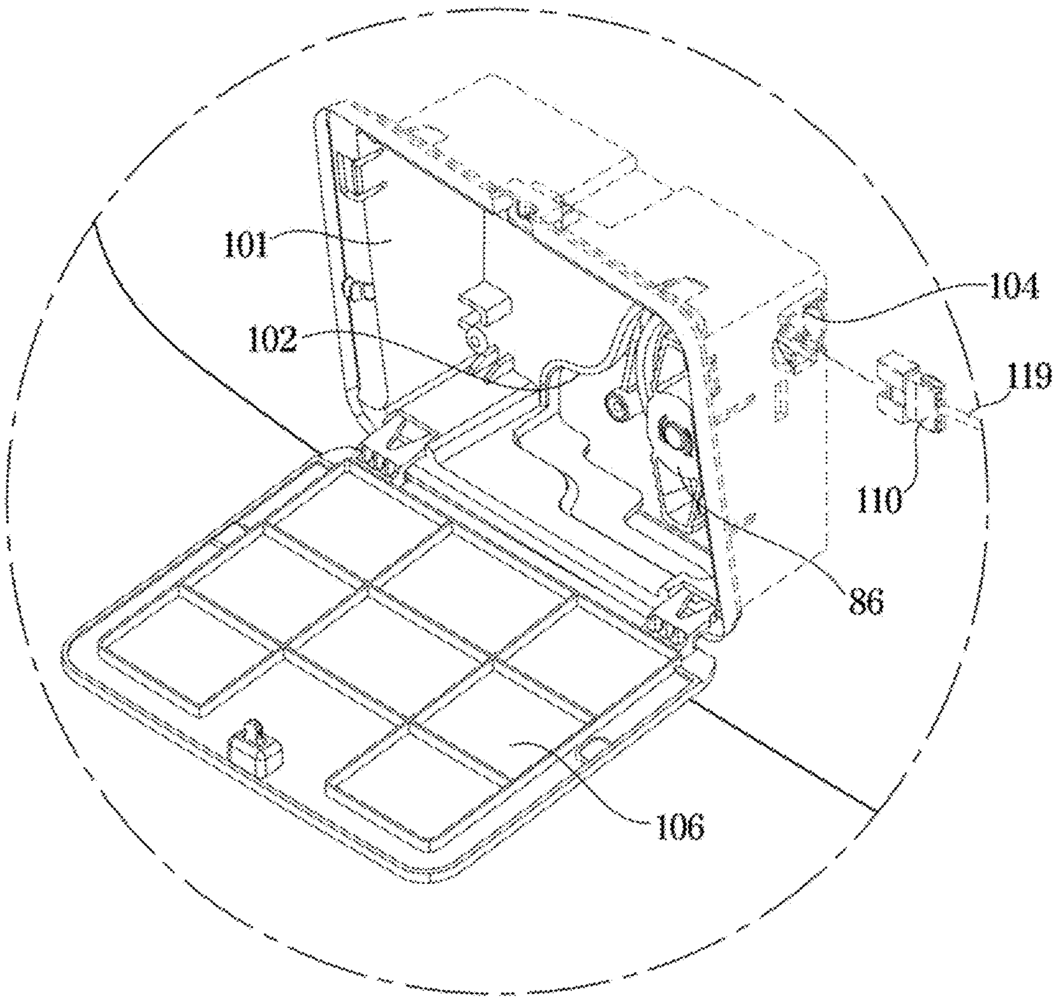


FIG. 9

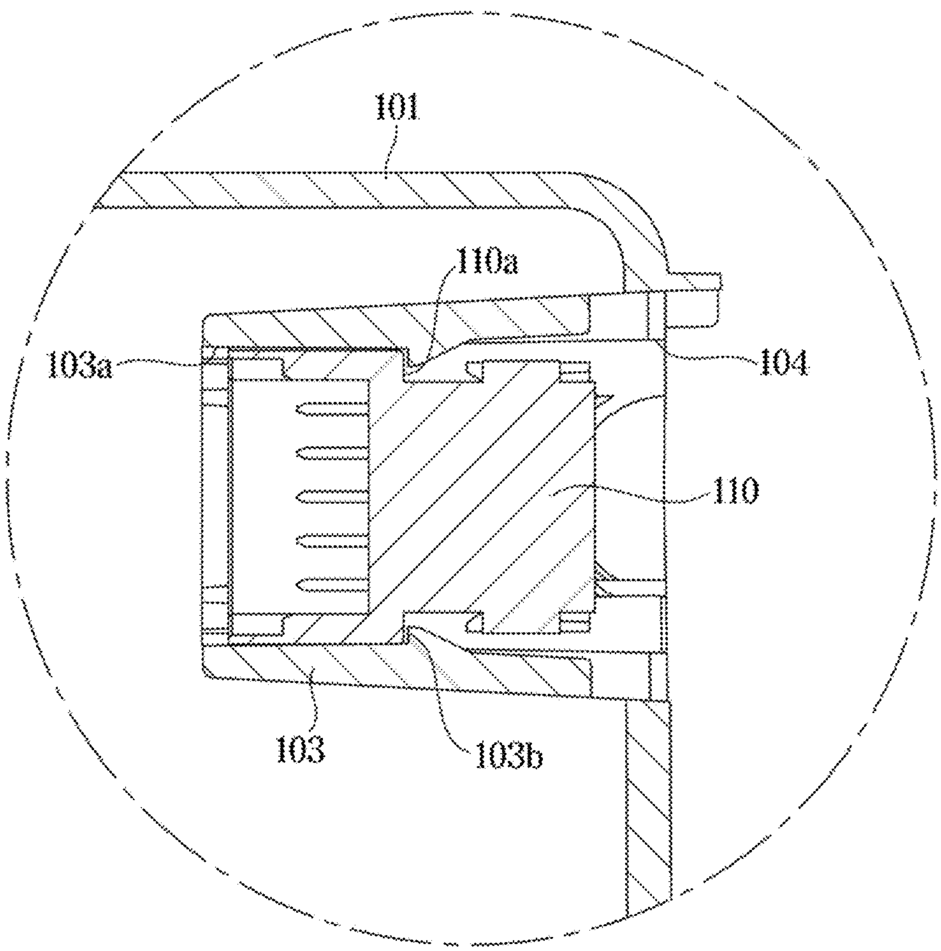


FIG. 10

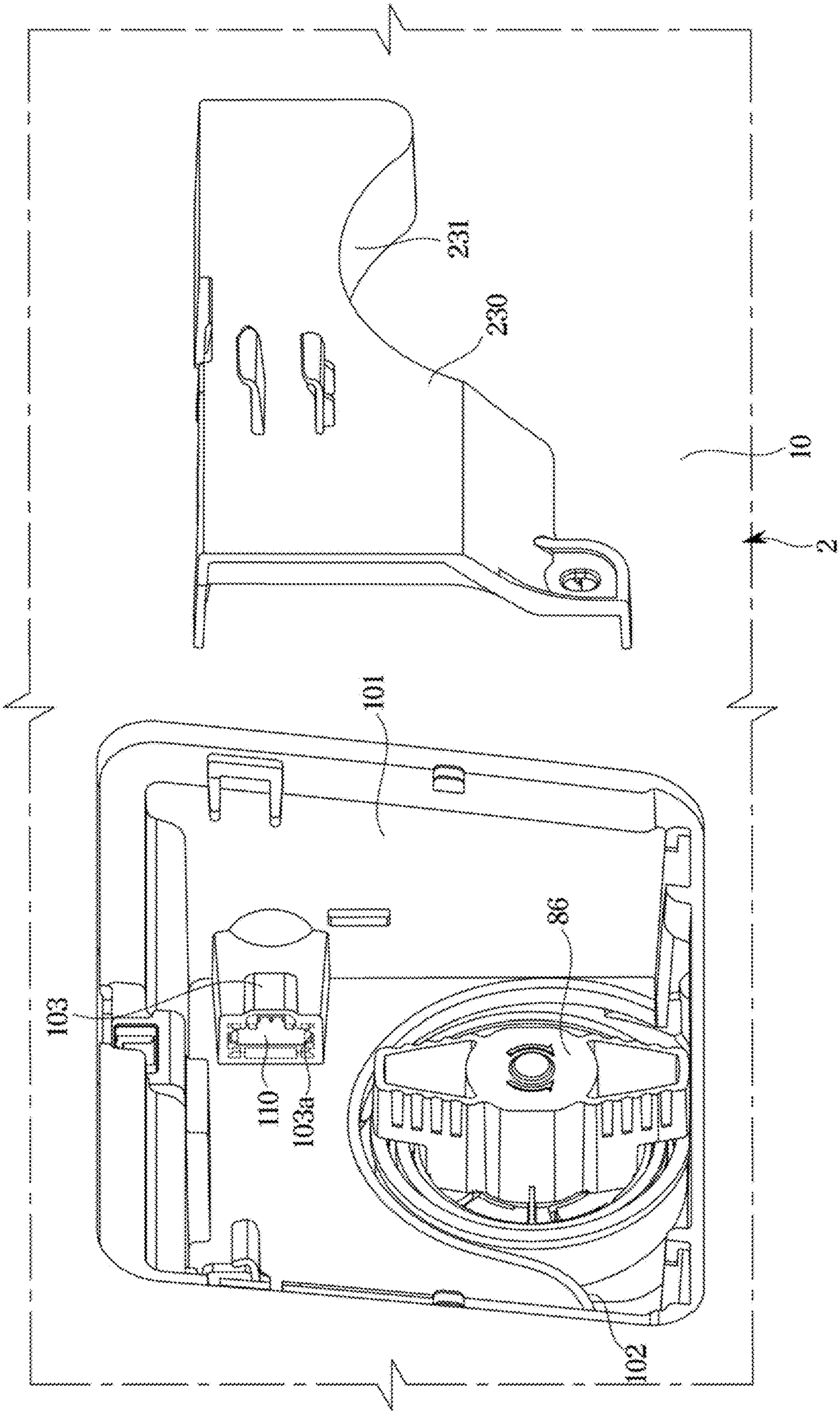


FIG. 11

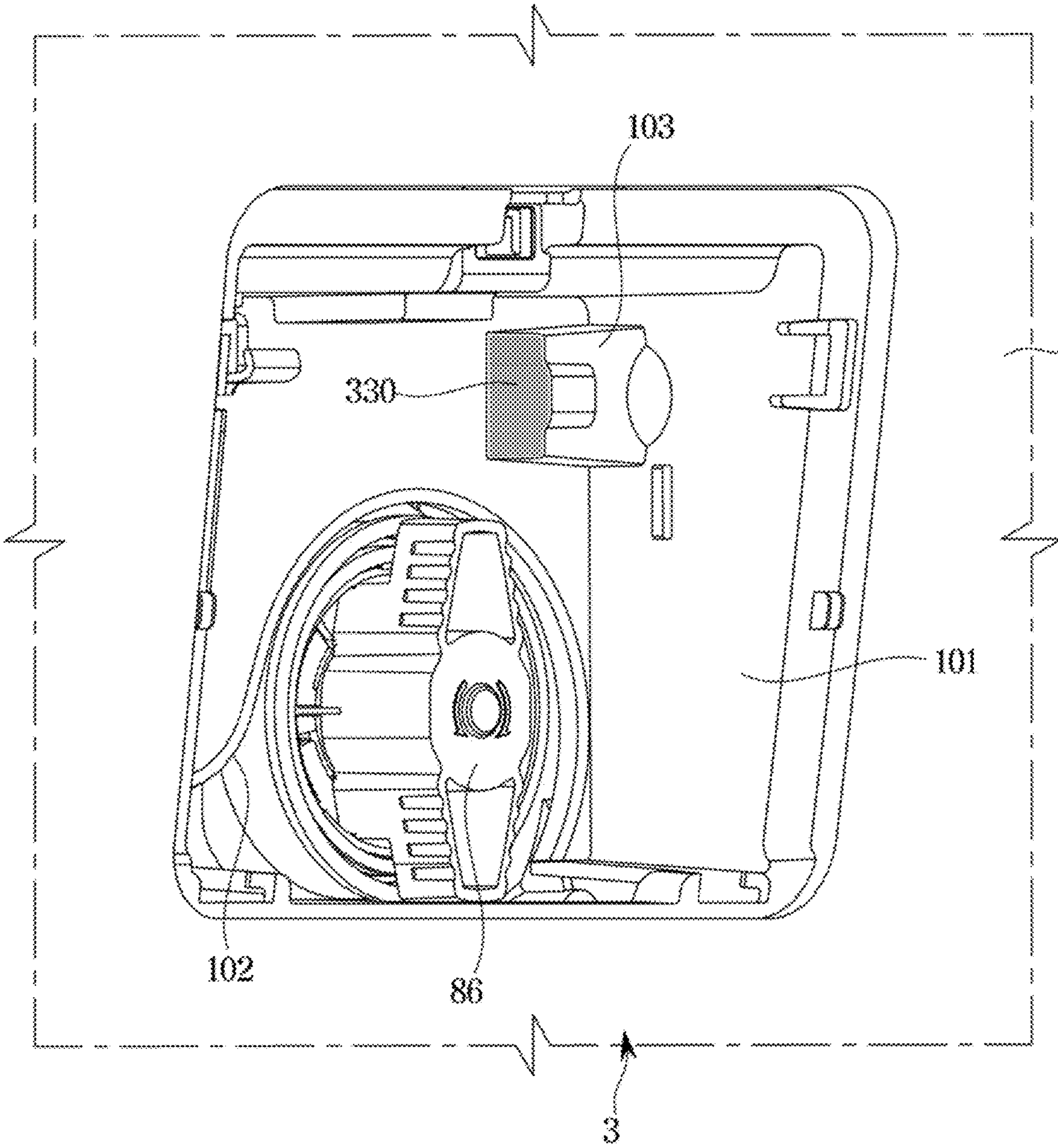


FIG. 12

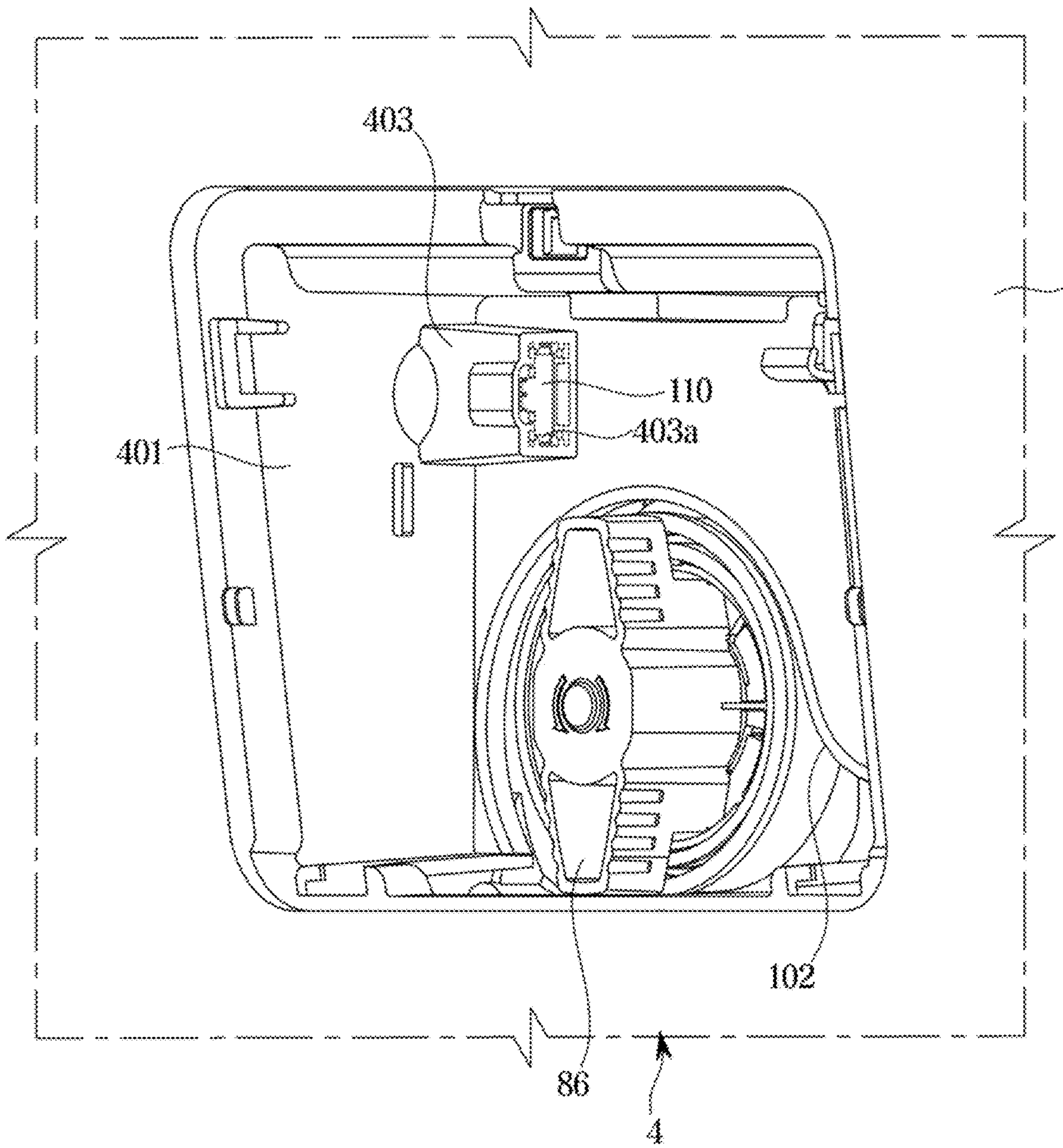


FIG. 13

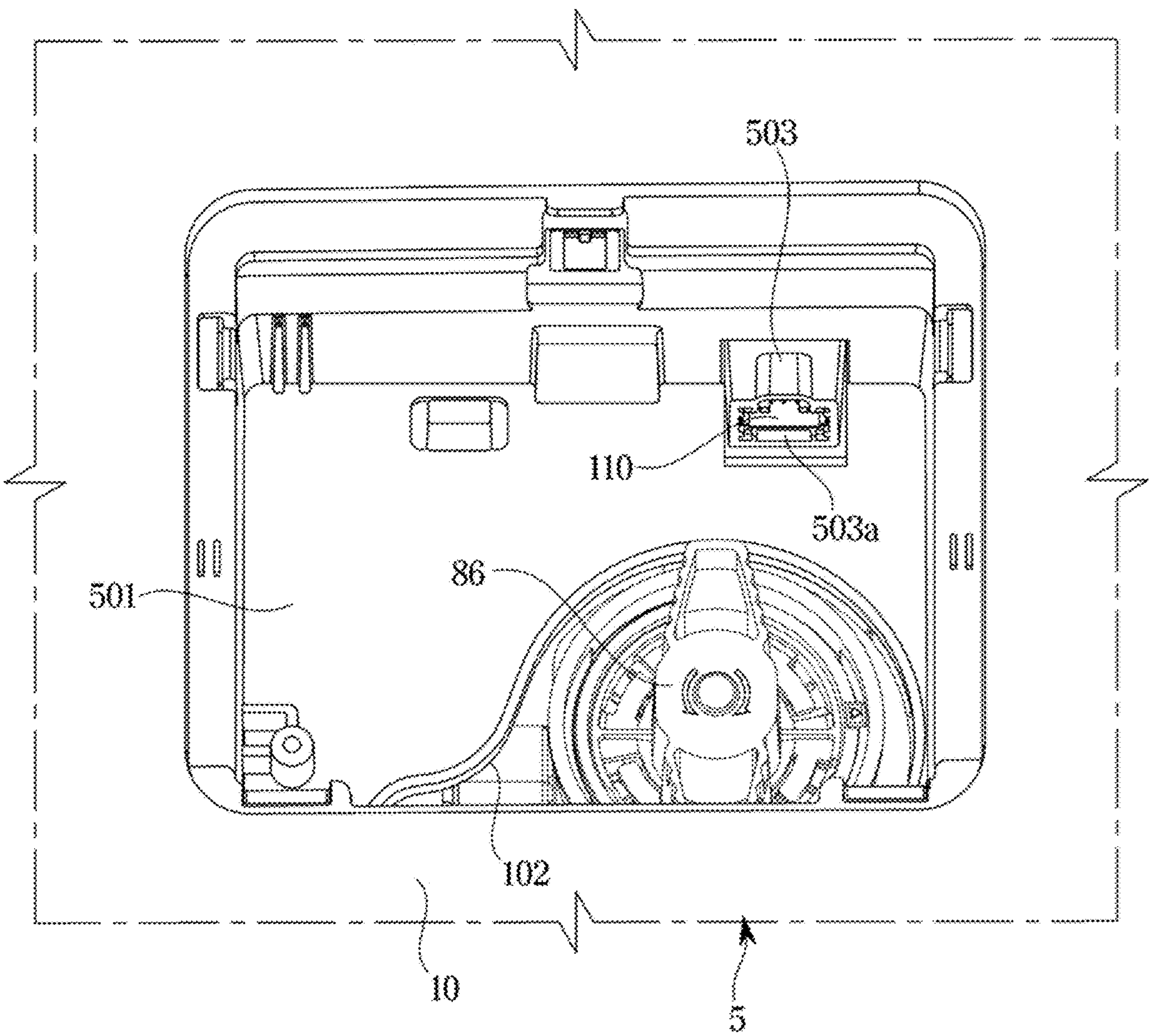
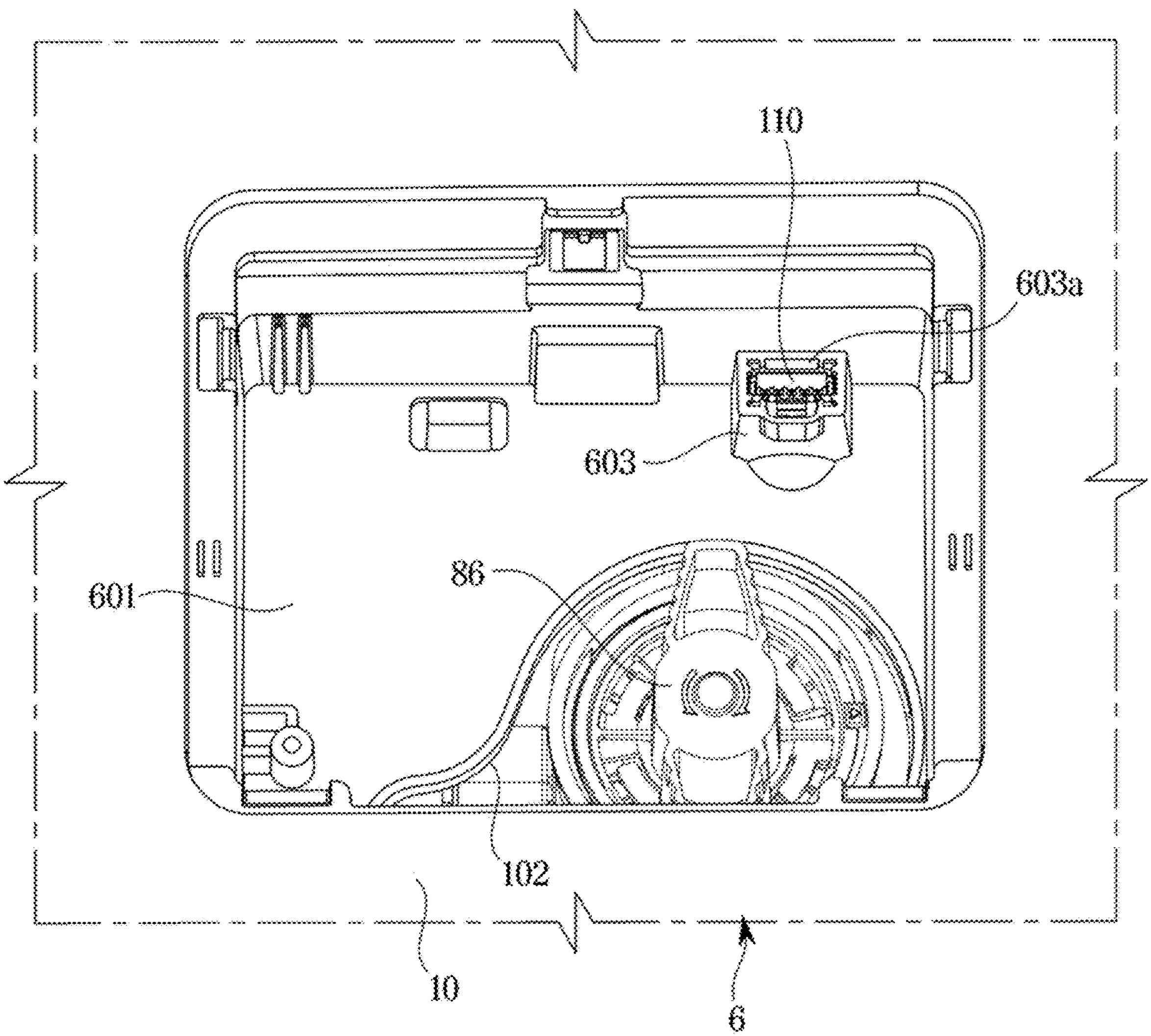


FIG. 14



WASHING MACHINE

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation application, under 35 U.S.C. § 111(a), of international application No. PCT/KR2022/001973 filed on Feb. 9, 2022, which claims priority to Korean Patent Application No. 10-2021-0018038, filed on Feb. 9, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference.

BACKGROUND

1. Field

The disclosure relates to a washing machine having a service port, and more specifically, to a washing machine having a drain pump and a service port.

2. Description of the Related Art

A washing machine is a household appliance that washes clothes, towels, bedding, and the like. Washing machines may be classified into drum type washing machines that rotate a drum and cause laundry to repeat rising and falling to wash the laundry and electronic type washing machines that wash laundry using a water flow generated by a pulsator when a drum rotates.

Regardless of the type of washing machine, operations performed by the washing machine may include a washing operation in which detergent water is supplied to a tub containing laundry and the laundry is washed while rotating a drum, a rinsing operation in which rinsing water is supplied to the tub and the drum is rotated to rinse the laundry, and a spin-drying operation in which water is discharged from the tub and the drum is rotated to remove moisture from the laundry.

The washing machine may include a controller configured for controlling the above-described operations, and a service port for checking update and/or error codes of the controller. Typically, the service port is located on a rear surface of the washing machine not to damage the appearance of the washing machine, for which it is difficult to access the service port.

SUMMARY

One aspect of the disclosure provides a washing machine with improved accessibility to a service port.

Another aspect of the disclosure provides a washing machine with improved accessibility to a service port without damaging the appearance thereof.

Another aspect of the disclosure provides a washing machine capable of preventing damage to a service port.

According to an aspect of the disclosure, there is provided a washing machine including: a cabinet; a tub disposed inside the cabinet; a drain pump configured to discharge water from the tub; a service port connectable to a service device and configured to receive or transmit error information of the washing machine therethrough when the service device is connected to the service port; and a pump housing mounted on a part of the cabinet in which the drain pump is located, wherein the service port is disposed inside the pump housing.

The washing machine may further include a pump cover detachably coupled to the pump housing to open or close the

pump housing, wherein when the pump cover is coupled to the pump housing the pump cover closes the pump housing so that the service port is hidden inside the pump housing.

When the pump cover covers the pump housing, the pump cover may form a part of a front surface of the cabinet.

The washing machine may further include a port cover detachably coupled to the pump housing to cover a part of the pump housing, and the pump cover is detachably coupled to the pump housing to cover the port cover and the pump housing.

The drain pump may include a pump handle, and the pump housing may include a housing opening through which the pump handle is exposed when the pump cover is open, and the port cover may include a cover opening to correspond to the housing opening so that the pump handle is exposed in the pump housing through the housing opening and the cover opening.

The pump housing may covers a part of the drain pump while the pump handle is exposed in the pump housing through the housing opening of the pump housing.

The pump housing may include a port mounting portion on which the service port is mounted, and the port mounting portion may include a connection opening to allow a part of the service port to be exposed.

The washing machine may further include a port cover to open and close the connection opening, and the port cover is made of an elastic material.

The port mounting portion may protrude from an inner surface of the pump housing that is covered by the pump cover.

The service port may be hook-coupled to the port mounting portion.

The service port may be accessible when the pump cover is open.

One end of the service port which is connected to the service device, may be located in a space between the tub and the pump housing.

The pump housing may include a material different from a material forming the cabinet.

The pump cover may be rotatably coupled to the pump housing to open or close the pump housing so that the service port is hidden inside the pump housing when the pump cover closes the pump housing.

The washing machine may further include a controller electrically connected to the service port, and when the service device is connected to the service port, the controller is configured to update firmware of the washing machine or transmit error codes included in the error information to the service device.

A washing machine including: a cabinet; a tub provided inside the cabinet; a drain pump configured to discharge water from the tub and having a pump handle; a pump housing mounted on a part of the cabinet in which the drain pump is located and configured to cover a part of the drain pump except the pump handle; a pump cover detachably coupled to the pump housing to cover the pump housing; and a service port arranged to protrude from one surface that is covered by the pump cover, and a port cover provided to prevent water from infiltrating into the service port.

The port cover may include a material having an elasticity.

The pump cover, when coupled to the pump housing, may form a part of a front surface of the cabinet while covering the service port.

The pump housing may be provided to separate a space in which the tub is located from a space the service port is located.

The pump housing may include a housing opening formed to allow the pump handle to be exposed when the pump cover is open, and the port cover may include a cover opening formed to correspond to the housing opening.

The service port may be provided to be accessible when the pump cover is open.

The washing machine may further include a controller electrically connected to the service port, and the controller, when the service port is connected to the service device, may be configured to update firmware or check error codes based on information transmitted from the service device.

As is apparent from the above, according to an aspect of the disclosure the washing machine has a service port located on a pump housing that is located on a front surface of a cabinet, so that access to the service port can be facilitated.

According to an aspect of the disclosure, the washing machine is provided such that when the service port is not in use, the service port is covered by a pump cover, so that accessibility to the service port is improved without damaging, the appearance of the washing machine.

According to an aspect of the disclosure, the washing machine includes a port cover that covers the service port, so that damage to the service port can be prevented.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a view illustrating a washing machine according to an embodiment of the disclosure.

FIG. **2** is a view illustrating a state in which a drain pump cover of the washing machine shown in FIG. **1** is open.

FIG. **3** is a view illustrating a side cross-section of the washing machine shown in FIG. **1**.

FIG. **4** is an enlarged view of a front surface of part A shown in FIG. **2**.

FIG. **5** is a front view illustrating a portion having a pump housing of the washing machine shown in FIG. **4**.

FIG. **6** is a view illustrating a service port located inside the pump housing shown in FIG. **5**.

FIG. **7** is a block diagram illustrating a connection relationship between a service device, a service port, and a controller when the service device is connected to the service port shown in FIG. **5**.

FIG. **8** is a view illustrating a state in which the service port shown in FIG. **4** is separated from the pump housing.

FIG. **9** is a view illustrating a part of a cross section taken along line B-B' of FIG. **4**.

FIG. **10** is a view illustrating a pump housing of a washing machine according to another embodiment of the disclosure.

FIG. **11** is a view illustrating a pump housing of a washing machine according to an embodiment of the disclosure.

FIG. **12** is a view illustrating a pump housing according to an embodiment of the disclosure.

FIG. **13** is a view illustrating a pump housing according to an embodiment of the disclosure.

FIG. **14** is a view illustrating a pump housing according to an embodiment of the disclosure.

DETAILED DESCRIPTION

The embodiments set forth herein and illustrated in the configuration of the disclosure are only the most preferred embodiments and are not representative of the full technical spirit of the disclosure, so it should be understood that they may be replaced with various equivalents and modifications at the time of the disclosure.

Throughout the drawings, like reference numerals refer to like parts or components.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to limit the disclosure. It is to be understood that the singular forms "a," "an," and "the" include plural references unless the context clearly dictates otherwise. It will be further understood that the terms "include", "comprise" and/or "have" when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

The terms including ordinal numbers like "first" and "second" may be used to explain various components, but the components are not limited by the terms. The terms are only for the purpose of distinguishing a component from another. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the disclosure. Descriptions shall be understood as to include any and all combinations of one or more of the associated listed items when the items are described by using the conjunctive term "~ and/or ~," or the like.

Hereinafter, embodiments according to the disclosure will be described in detail with reference to the accompanying drawings.

FIG. **1** is a view illustrating a washing machine according to an embodiment of the disclosure. FIG. **2** is a view illustrating a state in which a drain pump cover of the washing machine shown in FIG. **1** is open. FIG. **3** is a view illustrating a side cross-section of the washing machine shown in FIG. **1**.

Referring to FIGS. **1** to **3**, a washing machine **1** includes a cabinet **10** forming the external appearance, a tub **20** disposed in the cabinet **10**, a drum **30** rotatably disposed in the tub **20**, and a driving motor **40** to drive the drum **30**.

The cabinet **10** is provided at a front portion with an inlet **11** through which laundry may be inserted into the drum **30**. The inlet **11** is opened and closed by a door **12** installed on the front portion of the cabinet **10**.

A water supply pipe **50** to supply wash water to the tub **20** is installed at an upper portion of the tub **20**. One side of the water supply pipe **50** is connected to a water supply valve **56**, and the other side of the water supply pipe **50** is connected to a detergent box **52**.

The detergent box **52** is connected to the tub **20** through a connection pipe **54**. Water supplied through the water supply pipe **50** is supplied into the tub **20** together with a detergent via the detergent box **52**.

The tub **20** is supported by a damper **70**. The damper **70** connects an inner bottom surface of the cabinet **10** and an outer surface of the tub **20**.

The drum **30** includes a cylindrical portion **31**, a front panel **32** disposed in front of the cylindrical portion **31**, and a rear panel **33** disposed behind the cylindrical portion **31**. An opening for entry and exit of laundry is formed in the front panel **32**, and a shaft **90** to transmit power of the driving motor **40** is connected to the rear panel **33**.

A plurality of through holes **34** for distribution of wash water are formed around a circumference of the drum **30**, and a plurality of lifters **35** are installed on an inner peripheral surface of the drum **30** so that rising and falling of laundry may occur while the drum **30** is rotating.

The drum **30** and the driving motor **40** are connected through the shaft **90**, and according to the form of connection between the shaft **90** and the driving motor **40**, driving types may be classified into a direct driving type in which the shaft **90** is directly connected to the driving motor **40** to rotate the drum **30** and an indirect driving type in which a pulley **91** is connected between the driving motor **40** and the shaft **90** to drive the drum **30**.

The washing machine **1** according to one embodiment of the disclosure may be provided as the indirect driving type, but the disclosure is not limited thereto, and technical features of the disclosure may also be applied to the direct driving type.

One end of the shaft **90** is connected to the rear panel **33** of the drum **30**, and the other end of the shaft **90** extends to an outer side of a rear side portion **21** of the tub **20**. The other end of the shaft **90** may be provided to be inserted into the pulley **91** to obtain driving force from the driving motor **40**. In addition, a motor pulley **41** is formed on a rotating shaft of the driving motor **40**. A driving belt **92** may be provided between the motor pulley **41** and the shaft pulley **91** so that the shaft **90** may be driven by the driving belt **92**.

The driving motor **40** may be disposed at one side of a lower side portion of the tub **20** so that the driving belt **92** may drive the shaft **90** while rotating clockwise or counter-clockwise in the upper-lower direction of the tub **20**.

A bearing housing **96** is installed at the rear side portion **21** of the tub **20** so as to rotatably support the shaft **90**. The bearing housing **96** may be formed of an aluminum alloy and may be inserted into the rear side portion **21** of the tub **20** during injection molding of the tub **20**.

At a lower portion of the tub **20**, a drain pump **80** configured to discharge water inside the tub **20** to the outside of the cabinet **10**, a connecting hose **82** configured to connect the tub **20** and the drain pump **80** such that the water inside the tub **20** is introduced into the drain pump **80**, and a drain hose **84** configured to guide the water pumped by the drain pump **80** to the outside of the cabinet **10** are provided.

Meanwhile, a display **14** for displaying the state of the washing machine **1** to the user may be provided on a front upper portion of the cabinet **10**. The display **14** may include an inputter. A printed circuit board assembly (not shown) may be provided on the front upper portion of the cabinet **10**.

FIG. **4** is an enlarged view of a front surface of part A shown in FIG. **2**. FIG. **5** is a front view illustrating a portion having a pump housing of the washing machine shown in FIG. **4**. FIG. **6** is a view illustrating a service port located inside the pump housing shown in FIG. **5**. FIG. **7** is a block diagram illustrating a connection relationship between a service device, a service port, and a controller when the service device is connected to the service port shown in FIG. **5**. FIG. **8** is a view illustrating a state in which the service port shown in FIG. **4** is separated from the pump housing. FIG. **9** is a view illustrating a part of a cross section taken along line B-B' of FIG. **4**.

Referring to FIGS. **4** to **6**, the washing machine **1** may include a pump housing **101**, a pump cover **106** provided to cover the pump housing **101**, and a service port **110** mounted on the pump housing **101**.

The pump housing **101** may be mounted on the cabinet **10**. The pump housing **101** may cover the drain pump **80**. The pump housing **101** may cover the drain pump **80** while including a housing opening **102** configured to expose a pump handle **86**. The pump handle **86** exposed through the housing opening **102** is provided for an operator to maintain and/or repair the drain pump **80**.

The pump cover **106** may be provided to cover a front side of the pump housing **101** that is open. The pump cover **106** may cover to prevent the pump handle **86** exposed through the housing opening **102** from being exposed to the outside. The pump cover **106** may be detachably coupled to the pump housing **101**. The pump cover **106** may be detachably coupled to the cabinet **10**. When the pump cover **106** covers the open front side of the pump housing **101**, the pump cover **106** may form a part of the front surface of the cabinet **10**. Referring to FIG. **2**, the operator may move the pump cover **106** to an open position and may access the pump handle **86** of the drain pump **80**.

The service port **110** may be mounted on the pump housing **101**. The pump housing **101** may include a port mounting portion **103** provided to mount the service port **110** thereon. The port mounting portion **103** may be located on one surface of the pump housing **101** covered by the pump cover **106**. The port mounting portion **103** may be located on an inner surface of a right side of the pump housing **101**. The port mounting portion **103** may include a connection opening **103*a*** formed so that the service port **110** is connected to a service device **160**. A part of the service port **110** may be exposed through the connection opening **103*a***.

Referring to FIGS. **7** and **8**, the service port **110** may be electrically connected to a controller **150** by a port wire **119**. The part of the service port **110** exposed through the connection opening **103*a*** may be electrically connected to the service device **160**.

Referring to FIG. **8**, the pump housing **101** may include a coupling portion **104** formed on an outer surface of a left side thereof to be coupled to the service port **110**. The coupling portion **104** may be located at the other end of the port mounting portion **103** opposite to one end at which the connection opening **103*a*** is located.

Referring to FIGS. **8** and **9**, one end of the service port **110** connected to the service device **160** is disposed to face the connection opening **103*a***, and the other end of the service port **110** from which the port wire **119** extends is disposed to face the coupling portion **104**. The service port **110** may be hook-coupled to the port mounting portion **103**. The service port **110** may include a step portion **110*a***, and the port mounting portion **103** may include a port locking portion **103*b***. In a state in which the service port **110** is mounted on the port mounting portion **103**, the port locking portion **103*b*** supports the step portion **110*a*** and prevents the service port **110** from being separated from the port mounting portion **103**.

Referring to FIGS. **4** to **6**, when the pump cover **106** is in an open state, the operator may electrically connect the service device **160** to the service port **110** located on the pump housing **101**. The service device **160** connected to the service port **110** may be configured such that the controller **150** updates firmware of the washing machine **1** or checks and displays an error code of the washing machine **1**.

As described above, the washing machine **1** according to the embodiment of the disclosure has the service port **110** located in the pump housing **101** that is disposed on the front surface of the cabinet **10** rather than on the rear surface, so that accessibility to the service port **110** may be improved.

In addition, since the service port **110** is accessible from the front side of the washing machine **1**, the washing machine **1** does not need to be moved to access the service port **110**, so maintenance and/or repair of the washing machine **1** is facilitated.

In addition, since the service port **110** is generally covered by the pump cover **106** without being exposed to the outside, the appearance of the washing machine **1** may not be damaged.

FIG. **10** is a view illustrating a pump housing of a washing machine according to another embodiment of the disclosure.

Referring to FIG. **10**, a pump housing **101** of a washing machine **2** according to another embodiment of the disclosure will be described. The same reference numerals are assigned to the same components as those in the embodiment shown in FIGS. **4** to **6**, and detailed descriptions of the same components may be omitted.

Referring to FIG. **10**, the washing machine **2** according to the embodiment of the disclosure includes a port cover **230** that covers the service port **110**. The port cover **230** may be formed to cover a part of the pump housing **101** in which the service port **110** is disposed.

The port cover **230** may include a cover opening **231** formed to correspond to a housing opening **102**. As the cover opening **231** is formed to correspond to the housing opening **102**, the pump handle **86** may be exposed to the outside when the pump cover **106** is open.

The port cover **230** may be detachably coupled to the pump housing **101**. The port cover **230** may cover the service port **110** not to expose the service port **110** to the outside even when the pump cover **106** is open. The port cover **230** may be formed to prevent the service port **110** from being exposed to moisture. That is, the port cover **230** may be provided to prevent the service port **110** from being damaged by moisture. When the operator desires to update the firmware of the washing machine **2** or check the error code of the washing machine **2**, the operator may open the drain cover **106**, separate the port cover **230** from the pump housing **101**, and then access the service port **110**.

FIG. **11** is a view illustrating a pump housing of a washing machine according to an embodiment of the disclosure.

Referring to FIG. **11**, a pump housing **101** of a washing machine **3** according to another embodiment of the disclosure will be described. The same reference numerals are assigned to the same components as those in the embodiment shown in FIGS. **4** to **6**, and detailed descriptions of the same components may be omitted.

Referring to FIG. **11**, the washing machine **3** according to the embodiment of the disclosure may include a port cover **330** that covers the service port **110**. The port cover **330** may be provided to cover a connection opening **103***a* of the port mounting portion **103**. The port cover **330** may include a material having elasticity. The port cover **330** may be fitted to the connection opening **103***a* by elasticity to cover the service port **110**.

The port cover **330** may prevent moisture from penetrating into the service port **110**. The port cover **330** may prevent the service port **110** from being exposed to moisture or foreign substances. When the operator desires to update the firmware of the washing machine **3** or check the error code of the washing machine **3**, the operate may open the drain cover **106**, separate the port cover **330** from the pump housing **101**, and then access the service port **110**.

FIG. **12** is a view illustrating a pump housing according to an embodiment of the disclosure.

Referring to FIG. **12**, a pump housing **401** of a washing machine **4** according to an embodiment of the disclosure will be described. The same reference numerals are assigned to the same components as those in the embodiment shown in FIGS. **4** to **6**, and detailed descriptions of the same components may be omitted.

Referring to FIG. **12**, the pump housing **401** of the washing machine **4** according to the embodiment of the disclosure may include a port mounting portion **403** disposed on a left side inner surface thereof. Accordingly, the service port **110** may be inserted into the port mounting portion **403** from a left side outer surface of the pump housing **401**. The service port **110** may be exposed to the outside through the connection opening **403***a* of the port mounting portion **403** when the pump cover **106** is open.

FIG. **13** is a view illustrating a pump housing according to an embodiment of the disclosure.

Referring to FIG. **13**, a pump housing **501** of a washing machine **5** according to an embodiment of the disclosure will be described. The same reference numerals are assigned to the same components as those in the embodiment shown in FIGS. **4** to **6**, and detailed descriptions of the same components may be omitted.

Referring to FIG. **13**, the pump housing **501** of a washing machine **5** according to the embodiment of the disclosure may include a port mounting portion **503** disposed on an upper side inner surface thereof. Accordingly, the service port **110** may be inserted into the port mounting portion **503** from an upper side outer surface of the pump housing **501**. The service port **110** may be exposed to the outside through a connection opening **503***a* of the port mounting portion **503** when the pump cover **106** is open.

FIG. **14** is a view illustrating a pump housing according to an embodiment of the disclosure.

Referring to FIG. **14**, a pump housing **601** of a washing machine **6** according to an embodiment of the disclosure will be described. The same reference numerals are assigned to the same components as those in the embodiment shown in FIGS. **4** to **6**, and detailed descriptions of the same components may be omitted.

Referring to FIG. **14**, the pump housing **501** of the washing machine **5** according to the embodiment of the disclosure may include a port mounting portion **603** disposed on a rear side inner surface thereof. Accordingly, the service port **110** may be inserted into the port mounting portion **603** from an rear side outer surface of the pump housing **601**. The service port **110** may be exposed to the outside through a connection opening **603***a* of the port mounting portion **603** when the pump cover **106** is open.

Although few embodiments of the disclosure have been shown and described, the above embodiment is illustrative purpose only, and it would be appreciated by those skilled in the art that changes and modifications may be made in these embodiments without departing from the principles and scope of the disclosure, the scope of which is defined in the claims and their equivalents.

What is claimed is:

1. A washing machine comprising:

a cabinet;

a tub disposed inside the cabinet;

a drain pump configured to discharge water from the tub, the drain pump including a pump handle;

a service port connectable to a service device, and configured to receive or transmit error information of the washing machine therethrough in response to connecting the service device to the service port;

a pump housing in which the service port is disposed, and mounted on a part of the cabinet in which the drain pump is located, the pump housing including:

a housing opening through which the pump handle is exposed therethrough and a port mounting portion on which the service port is mounted; and

a pump cover detachably coupled to the pump housing to open or close the pump housing,

wherein:

the port mounting portion includes a connection opening to allow a part of the service port to be exposed in the pump housing;

in response to opening the pump housing by the pump cover, the pump handle and the service port are exposed and accessible;

in response to closing the pump housing by the pump cover, the pump cover covers the pump housing thereby covering the pump handle and the service port; and

the service port includes a step portion,

the port mounting portion includes a port locking portion; and

the port locking portion is configured to support the step portion to prevent the service port from being separated from the port mounting portion in response to mounting the service port on the port mounting portion.

2. The washing machine of claim **1**,

wherein in response to closing the pump housing by the pump cover, the service port is hidden inside the pump housing.

3. The washing machine of claim **2**, wherein in response to closing the pump housing by the pump cover, the pump cover forms a part of a front surface of the cabinet.

4. The washing machine of claim **2**, further comprising a port cover detachably coupled to the pump housing to cover a part of the pump housing where the service port is disposed within the pump housing, and the pump cover is detachably coupled to the pump housing to cover the port cover and the pump housing.

5. The washing machine of claim **4**, wherein the port cover includes a cover opening having a shape corresponding to a shape of the housing opening so that the pump handle is exposed in the pump housing while the port cover covers the part of the pump housing where the service port is located.

6. The washing machine of claim **5**, wherein the pump housing covers a part of the drain pump while the pump handle is exposed in the pump housing through the housing opening of the pump housing.

7. The washing machine of claim **1**, further comprising a port cover to open or close the connection opening, and the port cover is made of an elastic material.

8. The washing machine of claim **1**, wherein the port mounting portion protrudes from an inner surface of the pump housing that is covered by the pump cover.

9. The washing machine of claim **1**, wherein the service port is hook-coupled to the port mounting portion.

10. The washing machine of claim **2**, wherein the service port is accessible in response to opening the pump housing by the pump cover.

11. The washing machine of claim **1**, wherein the pump housing includes a material different from a material forming the cabinet.

12. The washing machine of claim **2**, wherein the pump cover is rotatably coupled to the pump housing to open or close the pump housing so that the service port is hidden inside the pump housing in response to closing the pump housing by the pump cover.

13. The washing machine of claim **1**, further including a controller electrically connected to the service port, and wherein in response to connecting the service device to the service port, the controller is configured to update firmware of the washing machine or transmit error codes included in the error information to the service device.

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